Digital Data Transmission 2026–27

Universitat Pompeu Fabra Barcelona

Modality A: Midterm exam (sample 3, key)

1. If $\phi(t)$ has energy E, then $\alpha\phi(t)$ has energy $|\alpha|E$ for any complex α.

(a) True
(b) False ⋆

> *Solution:* Statement 2. The energy of $\alpha\phi(t)$ is $|\alpha|^2 E$.

2. The representation $\bar{x}(t) = \sum_{i=1}^{N} x_i \phi_i(t)$ of a signal $x(t)$ on an orthonormal set is the orthogonal projection of $x(t)$ onto the signal space, the signal of the space closest to $x(t)$, and it equals $x(t)$ exactly when $x(t)$ belongs to the space.

(a) True ⋆
(b) False

> *Solution:* Statement 3, the representation as an orthogonal projection.

3. A rate or a capacity in bits per channel use is converted to bits per second by multiplying by the passband bandwidth W.

(a) True ⋆
(b) False

> *Solution:* Statement 5: a passband transmission has W channel uses per second.

4. Nyquist's criterion: with a baseband waveform of bandwidth W, which occupies $|f| \le W$, the minimum symbol period without inter-symbol interference is $T = \frac{1}{W}$.

(a) True
(b) False ⋆

> *Solution:* Statement 6. The maximum symbol rate is $2W$ symbols per second, so the minimum symbol period is $T = \frac{1}{2W}$.

5. If $x(t) = \sum_{i=1}^{N} x_i \phi_i(t)$ with $\{\phi_i\}_{i=1}^{N}$ orthonormal, its energy is the sum of the squared moduli of its coefficients, $E = \sum_{i=1}^{N} |x_i|^2$.

(a) True ⋆
(b) False

> *Solution:* Statement 9, energy of a signal from its coefficients on an orthonormal set.

6. If $x(t) = \sum_{i=1}^{N} a_i \phi_i(t)$ with unit-energy waveforms and independent zero-mean random coefficients, the average energy is $\mathrm{E}(E) = \sum_{i=1}^{N} \mathrm{E}(|a_i|^2)$ only when the waveforms are orthogonal.

(a) True
(b) False ⋆

> *Solution:* Statement 10. The average energy is $\sum_{i=1}^{N} \mathrm{E}(|a_i|^2)$ whatever the inner products of the waveforms, because the cross terms $\mathrm{E}(a_i a_k^*) = \mathrm{E}(a_i)\mathrm{E}(a_k)^*$ vanish.

7. In MATLAB, `randn` returns a real Gaussian sample of zero mean and unit variance, so σ `randn`$+\mu$ is Gaussian with mean μ and variance σ^2.

(a) True ⋆
(b) False

> *Solution:* Statement 12, Gaussian samples in MATLAB.

8. The binary entropy function $h_2(p)$ attains its maximum, 1 bit, at $p = 0$ and at $p = 1$.

(a) True
(b) False ⋆

> *Solution:* Statement 13. The binary entropy function vanishes at $p = 0$ and $p = 1$ and attains its maximum at $p = \frac{1}{2}$.

9. A coded-modulation system with a constellation of M points can transmit reliably above $\log_2 M$ bits per channel use when the channel has no noise.

(a) True
(b) False ⋆

> *Solution:* Statement 16. Even without noise, a constellation of M points cannot carry more than $\log_2 M$ bits per channel use.

10. The channel law of the complex AWGN channel, $W(y|x) = \frac{1}{\pi\sigma^2} e^{-|y-x|^2/\sigma^2}$, depends on y only through the distance $|y-x|$, so all outputs at the same distance from x are equally likely.

(a) True ⋆
(b) False

> *Solution:* Statement 18, the Gaussian channel law.

11. For two equiprobable real symbols $\pm A$ in real Gaussian noise of variance σ^2, the ML detector is a threshold at 0 and $P_{\mathrm{e}} = Q\left(\frac{A}{\sigma}\right)$, with $Q(u) = \int_u^\infty \frac{e^{-t^2/2}}{\sqrt{2\pi}}\,\mathrm{d}t$.

(a) True ⋆
(b) False

> *Solution:* Statement 23, error probability of binary antipodal signaling.

12. A repetition code of length r followed by an M-point constellation has rate $r \log_2 M$ bits per channel use.

(a) True
(b) False ⋆

Solution: Statement 25. The rate is $\frac{\log_2 M}{r}$: the constellation provides $\log_2 M$ bits per symbol and the repetition divides them by r.